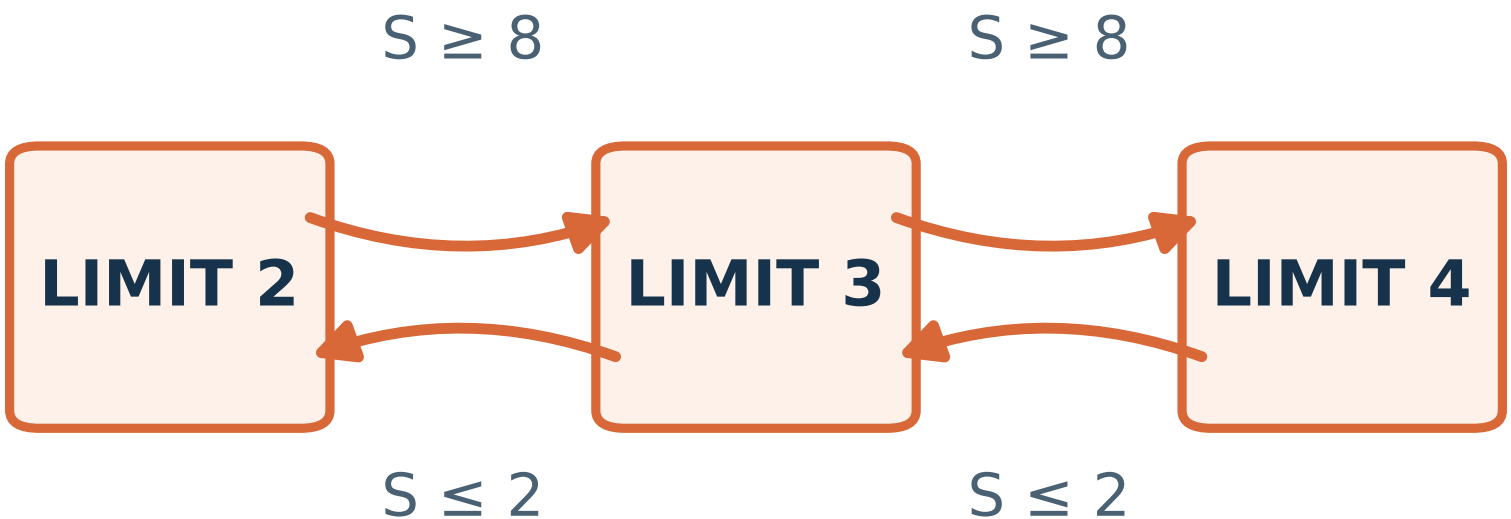
Update once per 32-cycle observation window



3 ≤ S ≤ 7: hold mode; limits are saturated at 2 and 4